

### 3.3.2 The Product Rule

**Motivation:** How might we go about finding the first derivative of  $f(x) = xe^x$ ?

**Fact:** In general

*Note: The next theorem will tell us how we should differentiate the product of two functions.*

**Theorem (The Product Rule):** Suppose that  $f$  and  $g$  are differentiable functions. Then,

*Proof:*

**Example 1:** Find the first derivative of  $f(x) = xe^x$ .

**Example 2:** Find the first derivative of  $f(t) = \sqrt{t}(3 - 6t)$

**Example 3:** Find the first derivative of  $f(y) = (y + 2\sqrt{y})e^y$ .

**Example 4:** Suppose that  $f$ ,  $g$ , and  $h$  are differentiable functions. Find a formula for  $\frac{d}{dx}[f(x)g(x)h(x)]$ .

**Example 5:** If  $f(x) = x^2 e^x \sqrt{x}$ , then find  $f'(x)$ .

**Example 6:** Suppose that  $f$  and  $g$  are differentiable functions and that  $f(x) = \sqrt{x} \cdot g(x)$ , where  $g(4) = 2$  &  $g'(4) = 3$ . Find  $f'(4)$ .